

Algebra 1
Unit 9
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Emma's work when solving the equation $x^2 - 8x = 21$ by completing the square is shown. Use Emma's work to answer questions 1 and 2.

$$x^2 - 8x = 21$$

$$(x^2 - 8x) - 21 = 0$$

$$(x^2 - 8x + 16) - 21 + 16 = 0$$

$$(x + 4)^2 - 5 = 0$$

$$(x + 4)^2 = 5$$

$$x + 4 = \pm\sqrt{5}$$

$$x + 4 = \sqrt{5} \text{ or } x + 4 = -\sqrt{5}$$

$$x = -4 + \sqrt{5} \text{ or } x = -4 - \sqrt{5}$$

$$x^2 - 8x = 21$$

$$(x^2 - 8x + 16) - 21 - 16 = 0$$

$$\sqrt{(x - 4)^2} - 37 = 0$$

$$(x - 4)^2 = 37$$

$$x - 4 = \pm\sqrt{37}$$

$$x = 4 \pm \sqrt{37}$$

1. Circle the step where Emma made an error.
2. Correct Emma's error and the steps that follow to find the correct solution(s) to the equation. Use the space provided above to show the correct steps.

Corbin's work when solving the equation $2x^2 - 12x = -7$ by completing the square is shown. Use Corbin's work to answer questions 3 and 4.

$$2x^2 - 12x = -7$$

$$2x^2 - 12x + 7 = 0$$

$$2(x^2 - 6x) + 7 = 0$$

$$2(x^2 - 6x + 9) + 7 - 9 = 0$$

$$2(x + 3)^2 - 2 = 0$$

$$2(x + 3)^2 = 2$$

$$(x + 3)^2 = 1$$

$$x + 3 = 1$$

$$x = -2$$

$$2(x^2 - 6x + 9) + 7 - 18 = 0$$

$$2(x - 3)^2 - 11 = 0$$

$$2(x - 3)^2 = 11$$

$$(x - 3)^2 = 5.5$$

$$x - 3 = \pm\sqrt{5.5}$$

$$x = 3 \pm \sqrt{5.5}$$

3. Circle the step where Corbin made an error.
4. Correct Corbin's error and the steps that follow to find the correct solution(s) to the equation. Use the space provided above to show the correct steps.

Solve each equation by completing the square. Show your work.

5. $x^2 + 2x - 3 = 21$

$$\begin{aligned}x^2 + 2x &= 24 \\x^2 + 2x + 1 &= 24 + 1 \\\sqrt{(x+1)^2} &= \sqrt{25} \\x + 1 &= \pm 5 \\x &= -1 + 5 \quad x = -1 - 5 \\x &= 4 \quad x = -6\end{aligned}$$

6. $x^2 + 12x - 4 = 50$

$$\begin{aligned}x^2 + 12x &= 54 \\x^2 + 12x + 36 &= 54 + 36 \\\sqrt{(x+6)^2} &= \sqrt{90} \\x + 6 &= \pm 3\sqrt{10} \\x &= -6 \pm 3\sqrt{10}\end{aligned}$$

7. $x^2 + 10x = -12$

$$\begin{aligned}x^2 + 10x + 25 &= -12 + 25 \\\sqrt{(x+5)^2} &= \sqrt{13} \\x + 5 &= \pm\sqrt{13} \\x &= -5 \pm \sqrt{13}\end{aligned}$$

8. $0 = \frac{3x^2 + 12x - 15}{3}$

$$\begin{aligned}x^2 + 4x - 5 &= 0 \\x^2 + 4x + 4 &= 5 + 4 \\\sqrt{(x+2)^2} &= \sqrt{9} \\x + 2 &= \pm 3 \\x &= 1 \quad x = -5\end{aligned}$$

9. $4x^2 = 48 - 16x$

$$\begin{aligned}\frac{4x^2 + 16x}{4} &= \frac{48}{4} \\x^2 + 4x &= 12 \\x^2 + 4x + 4 &= 12 + 4 \\\sqrt{(x+2)^2} &= \sqrt{16} \\x + 2 &= \pm 4 \\x &= -2 + 4 \\x &= -2 - 4 \\x &= 2 \\x &= -6\end{aligned}$$

10. $x^2 - 6x - 16 = 0$

$$\begin{aligned}x^2 - 6x &= 16 \\x^2 - 6x + 9 &= 16 + 9 \\\sqrt{(x-3)^2} &= \sqrt{25} \\x - 3 &= \pm 5 \\x &= 3 + 5 \\x &= 3 - 5 \\x &= 8 \\x &= -2\end{aligned}$$